



JavaScript Loops

Duration: 4 Hour

1-JavaScript Loops

Basic

1. Use a while loop to print numbers from 1 to 10 using while.

Example:

Output: 1 2 3 4 5 6 7 8 9 10

2. Use a for loop to iterate through an array and print each element.

Example:

Input: [1, 2, 3, 4, 5];

Output: 1 2 3 4 5

3. Use a for loop to print even numbers from 0 to 10.

Example:

Output: 0 2 4 6 8 10

4. Use a for loop to print the sum of numbers from 1 to 10.

Output: 55

5. Use a for loop to find the largest number in an array.

Example:

Input: [1, 2, 3, 4, 5];

Output: 5

6. Use a for loop to find the average of numbers in an array

Example:

Input: [1, 2, 3, 4, 5];

Output: 3

Mid



7. Use a for loop to find the factorial of a number.

Example:

Input: 5;
Output: 120

8. Use a for loop to print the Fibonacci sequence up to a given number.

Example:

Input: 10;
Output: 0 1 1 2 3 5 8

9. Use a for loop to print the prime numbers up to a given number.

Example:

Input: 20;
Output: 2 3 5 7 11 13 17 19

Advanced

10. Use a for loop to print the elements of a 2D array .

Example:

Input: [[1, 2, 3], [4, 5, 6], [7, 8, 9]];
Output: 1 2 3 4 5 6 7 8 9

11. Use a for loop to print the elements of an array in reverse order.

Example:

Input: [1, 2, 3, 4, 5];
Output: 5 4 3 2 1

12. Use a for loop to print the elements of an array with a specific step.

Example:

Input: [1, 2, 3, 4, 5], 2;
Output: 1 3 5



13. Use a for loop to find the frequency of a number in an array.

Example:

Input: [1, 2, 1, 3, 2, 1], 1;

Output: 3

14. Use the .map() method on the heros array to return a new array.

- The new array should rename the 'name' key to 'hero'.
- The 'name' key should not appear in the new array.
- The new array should have a new key added called (id).
- The key 'id' should be based on the index.

Example:

```
const heros = [  
  {name: 'Iron Man', power: 'Tech'},  
  {name: 'Spider-Man', power: 'Spider abilities'},  
  {name: 'Thor', power: 'Godly powers'},  
  {name: 'Hulk', power: 'Super strength'}  
]  
console.log(newHeros);  
/*  
[  
  {hero: 'Iron Man', power: 'Tech', id: 0},  
  {hero: 'Spider-Man', power: 'Spider abilities', id: 1},  
  {hero: 'Thor', power: 'Godly powers', id: 2},  
  {hero: 'Hulk', power: 'Super strength', id: 3}  
]
```

15. Write a JavaScript function that takes in the array "input Words" as an argument, and using the filter method, it returns a new array containing only the elements that have more than 7 characters and return the new Array

Example:

```
const inputWords = ["spray", "limit", "elite", "exuberant", "destruction", "present"];  
output: ["exuberant", "destruction", "present"]
```



16. Starting with an array containing the numbers 1 through 10, use reduce to produce for the following. Use console.log to display the results.

- The sum of the following: square the numbers divisible by 5.

- Example:
- `const numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10];`
- output
- `console.log(sumSquaredDivisibleBy5); // 625`